DEPRESSION DETECTION

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DECLARATION

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<u>CERTIFICATE</u>

This is to certify that the work titled **“Depression Detection”** submitted by, **Rishita Singh, Shinthil Singh, Shikhar Maheshwari** in partial fulfillment for the award of degree of **Bachelor of Technology Degree in Computer Science and Engineering & Information Technology** of Jaypee Institute of Information Technology, Noida has been carried out under my supervision. This work has not been submitted partially or wholly to any other University or Institute for the award of this or any other degree or diploma.

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SUMMARY

The developed system offers a complete pipeline for automated mental health risk detection using text from online social media, particularly Reddit. It combines both traditional machine learning models and large language model (LLM)-based methods. This allows for a clear comparison between classic supervised learning and modern generative AI techniques for assessing mental health.

At its core, the system processes raw Reddit posts through a structured preprocessing pipeline. This step removes unnecessary elements like URLs, special characters, and irrelevant content. It also includes normalization and filtering. Semantic representations of the cleaned text are created using sentence-transformer-based embeddings (MiniLM). These embeddings capture important context and language features in a compact vector format. They serve as input features for several classical machine learning models, including Logistic Regression, Support Vector Machines (SVM), and Multi-Layer Perceptron (MLP). These models provide a strong baseline for classifying mental health risk as high-risk or non-high-risk.

To go beyond basic supervised learning, the system uses large language models to test zero-shot and few-shot learning capabilities. Carefully designed prompt engineering strategies help the LLM-based approach determine if mental health risk can be inferred directly from text without specific training. Comparative experiments among zero-shot, few-shot, and trained models reveal the trade-offs in data efficiency, interpretability, and predictive performance.

The pipeline is modular and reproducible. It includes specific components for data preprocessing, embedding generation, classical model training, cross-validation, cross-domain transfer learning, prompt-based evaluation, and error analysis. Model performance is assessed using metrics like accuracy, precision, recall, and F1-score, along with confusion matrices to analyze misclassification patterns. Additionally, cross-domain experiments, such as between Reddit and DAIC, evaluate the system's ability to generalize across different data distributions.

Overall, this system acts as a unified experimental framework that compares traditional NLP-based machine learning models with new LLM-based methods. It emphasizes the strengths and weaknesses of each approach and offers valuable insights into scalable, automated mental health risk detection. This has promising applications in early intervention and digital mental health monitoring.

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CHAPTER 1: INTRODUCTION

1.1 Background

Depression and related mental health disorders are major global health issues that affect millions of people from various age groups and backgrounds. These conditions have a serious impact on quality of life and productivity. Traditional diagnostic methods depend on clinical interviews and self-reported assessments. These methods can be time-consuming, subjective, and hard to apply to large populations.

As social media use rises, people often share their emotions and mental states through online text. Platforms like Reddit have vast amounts of user-generated content discussing topics like depression, anxiety, and suicidal thoughts. This presents an opportunity to create automated systems that can detect early signs of mental health risks through text analysis.

Recent advances in artificial intelligence (AI) and natural language processing (NLP) allow us to identify meaningful patterns from this data. Sentence embedding models like MiniLM turn text into semantic vector representations that retain contextual information. Machine learning models such as Logistic Regression, Support Vector Machines (SVM), and Multi-Layer Perceptron (MLP) can use these embeddings for classification tasks.

This project suggests an automated system for detecting mental health risks that combines traditional machine learning with large language model (LLM)-based methods. We will also explore zero-shot and few-shot techniques using prompt engineering for comparison. The system aims to provide a scalable way to identify mental health risks and enable early identification and intervention.

1.2 Problem Definition and Objectives

The main challenge in this work is to automatically identify mental health risks from unstructured social media text. It also aims to maintain consistency and reliability across different user expressions. Unlike structured clinical assessments, online posts frequently use informal language, contain noise, and include subtle cues—like emotional distress, negative wording, or implied suicidal intent—which conventional models find hard to capture consistently. This makes detecting mental health risks a complicated issue that needs models to pull meaningful features from varied textual data.

Another challenge is to create an end-to-end process that cleans up noisy text, generates useful embeddings, and evaluates them using both traditional machine learning and LLM-based methods. Traditional models can have trouble generalizing across diverse data sets, while LLM-based approaches face issues with prompt design, consistency, and interpretability. These factors make it hard to develop a system that can accurately classify posts into different risk categories.

Current methods often depend on curated datasets or specific signals from the field, which limits their ability to scale to real-world social media data. Differences in writing style, context, and uneven class distribution also affect model performance, making it tough to detect mental health risks accurately and generally.

1.3 Significance/Novelty of the Problem

The significance of this problem lies in its potential for scalable mental health monitoring. Early detection of psychological risk can improve intervention outcomes and reduce long-term effects. Traditional diagnostic methods depend on clinical evaluations and self-reported assessments. These methods are time-consuming, subjective, and hard to access for large populations. By using social media text, the proposed system allows for continuous, low-cost analysis of mental health signals from real-world data. In places where mental health resources are limited, these automated systems can help connect individuals with timely support.

The novelty of this work comes from combining classical machine learning methods with large language model (LLM)-based techniques for detecting mental health risks. Many existing systems focus only on trained classifiers. This project looks at both supervised models and prompt-based methods, such as zero-shot and few-shot learning, within a unified framework. Additionally, using sentence-transformer embeddings ensures consistent meaning across all experiments. Including cross-domain evaluation and detailed error analysis further enhances the study by examining how well it generalizes and its practical use. This integrated approach provides a thorough comparison of traditional and modern NLP techniques for real-world mental health analysis.

1.4 Empirical Study (Field Survey/Existing Tool Study)

To put this work into context, an empirical survey of existing mental health detection methods was carried out, with an emphasis on the analysis of social media data through a text-based method. Previous approaches were mainly based on matching of the keywords, sentiment analysis and manually created linguistic features, whereas recent works are using transformer embeddings, like BERT or Sentence-BERT. Nevertheless, a great number of methods are struggling with generalization based on the differences in writing styles, noisy data, and platform dissimilarities.

Current applications like sentiment analysis systems or mental health surveys would offer a simple screening but would not suggest extended contextual insight into the reason why a user wants to be screened and emotional dimensions. They do not tend to pick up on implicit cues like distress patterns or suicidal ideation found in real-world text. Aspersory observations among students and young adults also show that there is a high level of preference to automated and available tools that can analyze everyday text without the need to have structured input or clinical data.

Previous studies indicate a moderate improvement of performance using advanced embeddings and deep learning models, but the majority of systems only consider trained classifiers and do not compare them with new LLM-based systems. Moreover, pipelines that combine preprocessing, embedding generation, classical models and prompt-based evaluation are seldom studied. Such shortcomings reveal the necessity of an overall system that integrates conventional machine learning with the current LLM methods to achieve more stable and scalable mental health risks detection.

1.5 Brief Description of the Solution Approach

The proposed solution is a modular mental health risk detection framework that integrates classical NLP-based machine learning with large language model (LLM)-based approaches. The system processes Reddit posts, extracts semantic embeddings, and evaluates them using a unified set of models. The core pipeline comprises:

- **Text Preprocessing:** Cleans social media text by removing URLs, mentions, special characters, and irrelevant content, followed by normalization and filtering.

● **Feature Extraction (MiniLM / Sentence-BERT):** Converts textual posts into dense semantic embeddings capturing contextual and emotional information.

● **Classical Models:** Logistic Regression, SVM, and MLP trained on embeddings to establish strong baselines for binary mental health risk classification.

● **LLM-based Evaluation:** Zero-shot and few-shot inference using prompt engineering to assess classification performance without explicit training.

The system is evaluated using stratified cross-validation to ensure robust performance estimation. For inference, a given post is pre-processed, converted into embeddings, and classified using trained models, while LLM-based predictions are generated through prompts for comparison. Additionally, cross-domain experiments are conducted to assess generalization across datasets. This integrated framework enables a comprehensive comparison between traditional supervised models and prompt-based LLM approaches for scalable mental health risk detection.

1.6 Comparison of Existing Approaches to the Problem

A number of solutions have tried to automate the process of identifying mental health through textual information with a specific focus on social media but most of them fail to extrapolate to the different writing styles and real-life situations. The initial techniques were based on manual features like n-grams, keyword search, or sentiment analysis with a moderate degree of accuracy without the ability to understand the context better. Logistic Regression or SVM models using TF-IDF features generally performed reasonably, but their performance declined on unseen or noisy data, indicating a lack of robustness.

More recent solutions employing deep-based learning and transformer-based models like BERT or Sentence-BERT achieved better contextual representation and classification. Yet, these models frequently need large labelled datasets and substantial computational resources, which are less practical to scale to. Moreover, the majority of available literature considers trained models only and fails to look at other types of inference, like prompt-based evaluation with large language models.

Conversely, the suggested framework offers a single comparison of classical machine learning frameworks (Logistic Regression, SVM, MLP) with those based on LLM (zero-shot and few-shot learning) on the basis of consistent MiniLM embeddings. This provides the opportunity to compare data-driven and prompt-based methods in the same pipeline. Moreover, there are cross-domain experiments and error analysis that are introduced to determine the evaluation of generalization and real-life applicability. It is an integrated solution that responds to the major shortcomings of current systems and is a step towards a more scalable and adaptable solution to automated mental health risk detection.

1.7 Extended Technology Background

Depression detection from online text is not a single model task, but a complete natural language processing workflow. The technology has to receive an informal post, remove unnecessary noise, preserve psychologically meaningful expressions, convert the language into numerical form, and then classify the post into a risk category. Each stage affects the final prediction. For example, a post may include casual spellings, repeated words, slang, emojis, links, quoted text, or very short phrases. A direct keyword search is therefore not enough because the emotional meaning of a sentence may be hidden in context rather than in one clear word.

The first technology layer is text preprocessing. It standardizes the input while keeping important mental health signals. URLs, excessive punctuation, and unrelated characters are removed because they do not usually help the classifier. At the same time, negation words, first-person expressions, references to hopelessness, loneliness, fear, exhaustion, and self-harm are preserved because they may carry risk-related information. Over-cleaning can reduce accuracy; therefore, preprocessing is designed to reduce noise without destroying the meaning of the original post.

The second layer is semantic representation. In the proposed system, MiniLM based sentence embeddings are used to transform each cleaned post into a dense vector. Unlike TF-IDF, which mainly counts words, embeddings capture the relationship between words and phrases. This is useful for depression detection because two posts may express similar distress using different vocabulary. For example, "I cannot keep going" and "I feel completely done" may not share many exact words, but their meanings are related. Sentence embeddings allow the model to recognize such similarity.

The third layer is classification. Logistic Regression, Support Vector Machine, and Multi-Layer

Perceptron are selected because they provide a balanced comparison between simplicity, margin-based decision making, and non-linear learning. These models are computationally efficient and can be trained on embeddings without requiring expensive end-to-end transformer fine-tuning. This makes the system practical for academic experiments as well as small-scale deployment.

1.8 Utility and Practical Relevance

The utility of the proposed system lies in its ability to support early screening rather than replace clinical diagnosis. In many situations, people express emotional distress online before they directly approach a counsellor, family member, or medical professional. A text-based detection system can help identify posts that may require attention and can support moderators, student wellness cells, counselling units, or research teams in prioritizing review. The system is therefore designed as an assistive technology, where the output is treated as a signal for further human evaluation.

A major practical advantage is scalability. Manual review of large volumes of social media text is difficult and time-consuming. Automated preprocessing and embedding generation allow thousands of posts to be represented consistently, while trained models can classify them quickly. The use of lightweight embeddings and classical machine learning reduces the requirement for high-end hardware. This makes the approach suitable for resource-constrained environments, including university laboratories and small research projects.

The hybrid comparison between classical machine learning and LLM-based methods also improves the utility of the study. Trained models provide more stable performance when labelled data is available, while zero-shot and few-shot LLM methods are useful when labelled examples are limited. Comparing both approaches in the same framework helps in understanding when supervised learning is preferable and when prompt-based reasoning may be useful. This is important for practical mental health applications, where labelled data is often scarce and ethically sensitive.

1.9 Data Characteristics and Risk Signals

Mental health related posts often contain subtle linguistic and emotional cues. Common indicators include negative self-references, expressions of isolation, sleep problems, loss of interest, guilt, hopelessness, and direct or indirect references to self-harm. However, these cues are not always explicit. Some users describe their condition indirectly through metaphors, humour, or fragmented writing. The system therefore needs semantic understanding rather than simple word matching. It must also avoid interpreting every sad or negative sentence as high risk, because temporary sadness and clinical distress are not identical.

Context is especially important. A post may mention suicide as a quotation, as support for another user, or while describing a past experience. These cases justify the inclusion of error analysis, prompt sensitivity analysis, and cross-domain testing in the proposed work.

CHAPTER 2 — LITERATURE SURVEY

2.1 Critical Review

Over the past decade, significant research has been conducted in the field of automated mental health detection using natural language processing (NLP) and machine learning techniques. Early studies primarily focused on traditional machine learning models such as Logistic Regression, Support Vector Machines (SVM), and Naïve Bayes using handcrafted features like TF-IDF, n-grams, and sentiment scores. These approaches achieved moderate performance but lacked the ability to capture contextual semantics effectively.

With the advancement of deep learning, researchers began utilizing neural networks such as LSTM and CNN for sequence modelling. Studies demonstrated that deep learning models improve performance by capturing temporal and contextual dependencies in text data. For example, deep neural networks have shown strong capability in detecting suicidal ideation from social media posts with improved predictive accuracy.

The introduction of transformer-based architectures such as BERT and RoBERTa further enhanced performance in depression detection tasks. These models provide contextual embeddings and outperform traditional methods by capturing deeper semantic relationships. Research combining BERT with Bi-LSTM has shown improved detection of depression and suicidal intent in Reddit data.

More recent work focuses on large language models (LLMs) such as ChatGPT and BART for zero-shot and few-shot classification. These models demonstrate strong potential for mental health detection without requiring labelled training data.

Additionally, explainable AI techniques are being integrated to improve interpretability of predictions. Methods such as LIME help identify linguistic patterns associated with depression, enhancing trust in AI-based systems.

Despite these advancements, challenges remain in generalization, data scarcity, and domain variability. Many studies rely on specific datasets and fail to perform well across different domains.

Overall, the literature shows a clear progression:

- Traditional ML → Deep Learning → Transformers → LLMs
- Increasing focus on **context, generalization, and scalability**

2.1.1 Detecting Depression from social media Text using Machine Learning

Category	Details
Reference	Coppersmith, G., Dredze, M., & Harman, C. (2014). Quantifying mental health signals in Twitter. ACL Workshop.
Description	Uses machine learning to detect depression from social media posts using linguistic patterns.
Feature	TF-IDF features, n-grams, and linguistic indicators.
Dataset Used	Twitter dataset (mental health labeled users).
Accuracy	~70%

Table 2.1 Analysis of Paper 1

2.1.2 Predicting Depression via Social Media

Category	Details
Reference	De Choudhury, M., Counts, S., & Horvitz, E. (2013). Predicting depression via social media. ICWSM.
Description	Predicts depression using behavioral and linguistic signals from social media activity.
Feature	Posting frequency, sentiment, linguistic style.
Dataset Used	Twitter dataset
Accuracy	Moderate

Table 2.2 Analysis of Paper 2

2.1.3 Depression Detection using Machine Learning

Category	Details
Reference	Nguyen, T., Nguyen, T., & Pham, T. (2020). Depression detection from social media text using ML.
Description	Uses classical ML models for depression detection.
Feature	TF-IDF + sentiment features
Dataset Used	Social media dataset
Accuracy	~75%

Table 2.3 Analysis of Paper 3

2.1.4 Sentence-BERT for Semantic Text Representation

Category	Details
Reference	Reimers, N., & Gurevych, I. (2019). Sentence-BERT. EMNLP.
Description	Introduces sentence embeddings for semantic similarity and classification tasks.
Feature	Transformer-based sentence embeddings
Dataset Used	General NLP datasets
Accuracy	High (task-dependent)

Table 2.4 Analysis of Paper 4

2.1.5 BERT-based Depression Detection

Category	Details
Reference	Various (2023). BERT-based approach for depression detection.
Description	Uses BERT for contextual classification of mental health text.
Feature	Contextual embeddings
Dataset Used	Reddit / Twitter
Accuracy	~80–85%

Table 2.5 Analysis of Paper 5

2.1.6 LSTM-based Mental Health Detection

Category	Details
Reference	Various (2020). LSTM for depression detection.
Description	Uses sequential modeling for text classification.
Feature	Word embeddings + sequence modeling
Dataset Used	Social media text
Accuracy	~78%

Table 2.6 Analysis of Paper 6

2.1.7 Multimodal Depression Detection

Category	Details
Reference	DAIC-WOZ based studies
Description	Combines text, audio, and behavioral features.
Feature	Multimodal fusion
Dataset Used	DAIC-WOZ
Accuracy	~85%

Table 2.7 Analysis of Paper 7

2.1.8 Zero-shot Learning using LLMs

Category	Details
Reference	Brown et al. (2020). GPT-3
Description	Introduces zero-shot and few-shot learning using LLMs.
Feature	Prompt-based inference
Dataset Used	General NLP tasks
Accuracy	Task-dependent

Table 2.8 Analysis of Paper 8

2.1.9 BART for Zero-shot Classification

Category	**Details**
Reference	Facebook AI (BART model)
Description	Uses NLI-based classification for zero-shot tasks.
Feature	Sequence-to-sequence transformer
Dataset Used	Multi-domain
Accuracy	Moderate (~55–70%)

Table 2.9 Analysis of Paper 9

2.2 Summary Table

Paper Type	Method	Accuracy	Strength	Limitation
Traditional ML	TF-IDF + LR/SVM	~70–75%	Simple	No context
Deep learning	LSTM/CNN	~78%	Sequence modeling	Needs data
Transformers	BERT/SBERT	~80–85%	Strong context	High cost
Multimodal	Text+Audio	~85%	High accuracy	Complex
LLM	Zero-shot	~55–70%	No training	Prompt dependent

Table 2.10 Summary table

2.3 Research Gaps

1. No Unified Comparison

Most works evaluate only one approach (ML or DL or LLM).

2. Lack of Prompt Engineering Analysis

LLM studies rarely explore:

- Prompt sensitivity

- Few-shot vs zero-shot

3. Limited Cross-Domain Testing
Models fail when applied to different datasets.

4. Heavy Computational Requirements
Transformer models are not resource-efficient.

5. Lack of Error Analysis
Most studies focus only on accuracy.

2.4 Extended Literature Survey on Technology Trends

The literature on automated depression detection shows a gradual movement from surface-level text analysis to context-aware language understanding. Initial approaches mainly used lexicons, keyword lists, sentiment scores, n-grams, and TF-IDF features. These methods were attractive because they were simple, interpretable, and easy to implement using traditional classifiers. Studies such as social media based depression prediction and mental health signal quantification demonstrated that linguistic patterns in user posts can provide measurable indicators of mental health risk. However, the main limitation of such approaches is that they depend heavily on the exact words used by the writer.

In keyword-based systems, phrases such as "depressed", "hopeless", or "suicidal" may directly increase the risk score. Although this is useful for clear cases, it fails when users describe distress indirectly. Many people avoid clinical vocabulary and instead use expressions such as "I feel empty", "I cannot sleep", or "nothing feels worth it". Statistical features such as TF-IDF improve over pure keyword search because they consider a larger vocabulary, but they still do not fully represent semantic similarity between different expressions.

Traditional machine learning models such as Logistic Regression, Naive Bayes, and Support Vector Machines were widely used with these features. Their advantage is that they are fast and relatively transparent. Logistic Regression can show which features influence the prediction, while SVM can create a strong decision boundary in high-dimensional text space. Nevertheless, their performance depends strongly on feature engineering. When the writing style, platform, or topic changes, the learned word patterns may not generalize well.

This limitation created the need for representation learning. Instead of manually deciding which words are important, newer methods learn numerical representations from large corpora. Word embeddings, sentence embeddings, and transformer embeddings allow models to capture meaning at different levels. This progression is central to the proposed project, which uses sentence-level embeddings to represent complete posts rather than isolated terms.

2.4.1 Deep Learning and Sequence Based Approaches

Deep learning methods introduced an important shift in mental health text classification by modeling word order and sequential dependencies. Recurrent models such as LSTM are able to process a sentence from beginning to end and retain information about previous words. This is useful when the emotional meaning of a sentence depends on the relationship between multiple phrases. For example, the sentence "I thought I was getting better, but now everything feels worse" contains a contrast that simple bag-of-words methods may not represent effectively.

Convolutional neural networks have also been used for text classification by detecting local phrase patterns. They can identify combinations of words that frequently occur in high-risk posts. However, both CNN and LSTM models usually require substantial labelled data and careful tuning. They may also overfit when the dataset is small or when the class distribution is imbalanced. In mental health applications, this is a practical difficulty because reliable labelled data requires ethical collection, expert annotation, and privacy protection.

Another challenge in deep learning based depression detection is interpretability. Although neural models may improve accuracy, their internal decision process can be difficult to explain. In a sensitive domain such as mental health, a prediction should ideally be accompanied by some understanding of why the post was flagged. This is one reason why the present work retains classical models along with embeddings: the representation is modern, but the classifiers remain efficient and easier to compare.

2.3.1 Transformer Based Representations

Transformer architectures further improved the field by using attention mechanisms to identify relationships between words regardless of their position in the sentence. BERT, RoBERTa, Sentence-BERT, and related models generate contextual representations, meaning that the vector for a word or sentence changes depending on surrounding text. This is valuable for mental health analysis because the same word can have different meanings in different contexts. For example, "tired" may describe ordinary fatigue in one post and severe emotional exhaustion in another.

2.3.2 Sentence Embeddings and MiniLM

Sentence-BERT and MiniLM based approaches are particularly relevant for this project because they represent the entire post as a compact semantic vector. MiniLM is designed to be lightweight while retaining strong language understanding. It is faster and less resource-intensive than large transformer models, making it suitable for repeated experimentation, cross-validation, and CPU-based execution. This creates a practical middle path between older TF-IDF features and expensive full transformer fine-tuning.

Sentence embeddings also simplify the experimental pipeline. Once embeddings are generated, the same feature vectors can be used with different classifiers. This makes it possible to compare Logistic Regression, SVM, and MLP under consistent conditions. It also reduces the chances that performance differences are caused by different preprocessing or feature extraction methods. For a research project, this consistency is important because it allows fair comparison across models.

2.3.3 LLM Based and Prompt Based Methods

Recent literature has explored large language models for zero-shot and few-shot classification. Instead of training a model on a labelled dataset, a prompt describes the task and asks the model to classify the input text. This is attractive in low-data environments because it reduces the need for task-specific training. Models such as GPT-style systems and BART-based zero-shot classifiers can use general language knowledge to make predictions on new tasks.

However, LLM-based methods introduce new challenges. Their performance depends strongly on prompt wording, label descriptions, and the examples provided in few-shot settings. A small change in the prompt can change the final prediction. In safety-critical domains, this instability is important. The literature therefore suggests that LLM outputs should be evaluated carefully rather than accepted as automatically reliable. The proposed project addresses this issue by including prompt sensitivity analysis and comparing zero-shot and few-shot outcomes with trained models.

Another concern is calibration. An LLM may produce a confident prediction even when the text is ambiguous. For mental health risk detection, this can be risky because false negatives may delay intervention and false positives may cause unnecessary concern. Therefore, LLM methods are best treated as complementary tools within a broader pipeline rather than as standalone decision makers.

2.3.4 Cross-Domain Generalization in Existing Work

A repeated issue in the literature is domain shift. Models trained on one platform may not perform equally well on another platform because users write differently across communities. Twitter posts are short and often contain hashtags, while Reddit posts are usually longer and more discussion-oriented. Clinical interview datasets such as DAIC-WOZ contain structured conversation rather than open social media text. These differences affect vocabulary, sentence length, topic distribution, and emotional expression.

Many studies report strong results on a single dataset but provide limited evidence about cross-domain performance. This makes it difficult to judge whether the model has learned general mental health signals or only dataset-specific patterns. Cross-domain testing is therefore necessary for practical deployment. The inclusion of Reddit and DAIC transfer experiments in the present work responds directly to this gap by measuring how performance changes when the model is applied outside its original training distribution.

2.4 Consolidated Insights from Literature

The reviewed literature suggests five important conclusions. First, linguistic content on social media contains useful signals for depression and risk detection. Second, traditional machine learning is efficient but limited by surface-level features. Third, deep learning and transformer models improve contextual understanding but can require more data and computation. Fourth, LLM-based methods are useful in low-data settings but are highly sensitive to prompt design. Fifth, most existing studies need stronger generalization analysis and clearer error interpretation.

The proposed project is positioned at the intersection of these findings. It uses semantic embeddings to improve representation, classical models to maintain efficiency, LLM-based methods to study prompt-driven inference, and cross-domain testing to examine robustness. The literature survey therefore supports the need for a unified framework that compares multiple approaches under the same evaluation setting.

CHAPTER 3: System Design and Methodology

3.1 Design Consideration

The various practical, ethical, and technical factors in the development of the proposed mental health risk detection system will help in ensuring that the solution is scalable, reliable, and socially responsible.

Socially and ethically, the system is associated with sensitive user-generated content associated with mental health. Thus, privacy and anonymity are paramount. The publicly available dataset (Reddit posts) is not explicitly handled with personally identifiable information. The system is not intended to diagnose but assertively to analyze, having the outputs being interpreted as indicators and not as definitive judgments. Moral issues like misclassification are also addressed because a false positive or false negative result of mental health detecting can be disastrous. Therefore, the evaluation metrics such as recall and F1-score have been considered as a priority to balance the sensitivity and precision.

Regarding health and safety, the system is to facilitate early identification of risky mental conditions, e.g. suicidal thoughts. Although it is not a substitute of medical personnel, it can serve as a first line screening device to indicate potentially worrisome content. The design guarantees the interpretability of outputs and their review by expert in case of deployment in real-life situations.

Technically, important considerations are scalability and generalization. The data of social media is extremely noisy and heterogeneous, which needs powerful methods of preprocessing and feature extraction. The system relies on sentence embeddings (MiniLM) which are able to capture semantic meaning in an efficient manner and are computationally feasible. Also, the modular architecture enables each of the components, including preprocessing, embedding, classification, and evaluation, to be optimized separately.

Environmental and economic factors are considered, with the choice of lightweight models and efficient embedding methods. The system, rather than training with large, resource-intensive models of transformers, trains with pre-trained embeddings, with classical machine learning models, making it cheaper to train. Approaches based on LLM are applied controllably (zero-shot/few-shot) without a lot of fine-tuning, and are even more resource-efficient.

Lastly, the system is extensible. It can be easily integrated with new datasets, improved embeddings, or advanced models without restructuring the whole pipeline. This will guarantee long-term applicability and flexibility to the changing trends in mental health analysis research.

3.2 Methodology

The suggested system adheres to a systematic pipeline in identifying mental health risk based on Reddit text data. The model is broken down into several steps, whereby there is a clear flow of raw data to final assessment.

Step 1: Data Collection

The data on mental health (depression, anxiety, and suicide watch) communities on Reddit is obtained as text data. The content of the post and metadata such as subreddit labels can be found in each record.

Step 2: Text Preprocessing

Noisy words like URLs, mentions and special characters are filtered out of raw text. Text normalization (lowercase, whitespace treatment) is used. Duplicates or useless posts are filtered off, and quality input is obtained to be further processed.

Step 3: Feature Extraction

MiniLM sentence embeddings are used to convert the cleaned text to numerical form. These embeddings are semantically and contextually rich, allowing them to represent better than the conventional algorithms, such as TF-IDF.

Step 4: Model Training

Three classical models are trained:

Logistic Regression (baseline)

Support Vector Machine (high performance)

Multi-Layer Perceptron (non-linear learning)

Step 5: LLM-based Evaluation

Large language models are used to apply zero-shot and few-shot prompting techniques. This enables the comparison of trained models to inference without explicit training based on prompts.

Step 6: Evaluation

The models are tested on:

Accuracy

Precision

Recall

F1-score

Misclassifications can also be analyzed by using confusion matrices.

Step 7: Cross-Domain Testing

To test generalization ability, models trained on one dataset are tested on a different dataset.

This approach will guarantee a thorough comparison of both the conventional machine learning and the current approaches based on LLM.

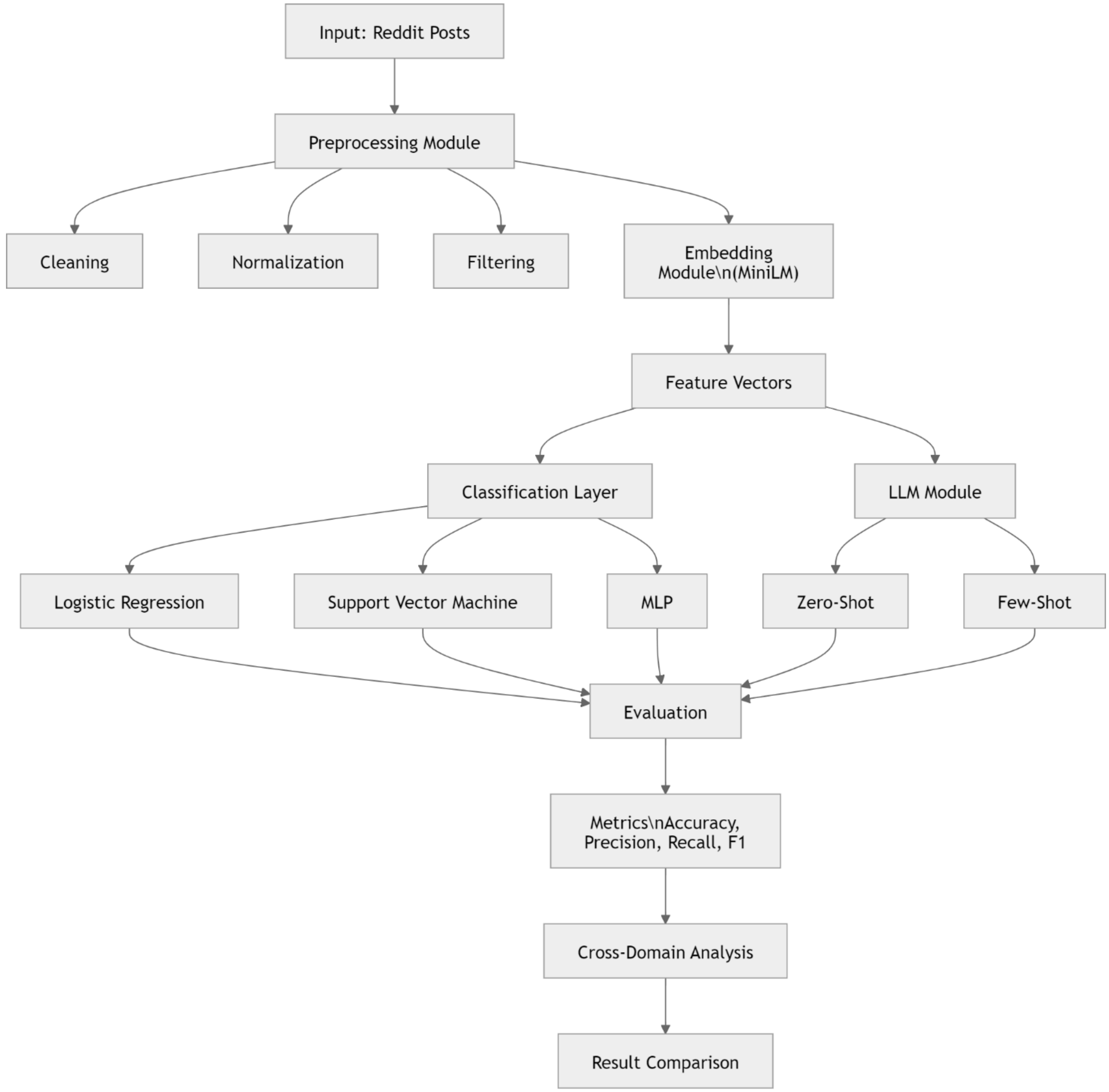

Fig 3.1 System Architecture

3.3 Modelling

The modelling component focuses on transforming textual data into meaningful representations and applying classification algorithms to predict mental health risk.

Mathematical Representation

Text input is represented as:

$$T = \{t_1, t_2, \dots, t_n\}$$

Each text sample is converted into an embedding vector:

$$X = f(T)$$

where f is the MiniLM embedding function.

The classification task is defined as:

$$y = g(X)$$

where g represents the machine learning model.

Using Embedding Function

$$X = f(T)$$

Classification Function

$$y = g(X)$$

Models Used

- **Logistic Regression:** Linear decision boundary
- **SVM:** Maximizes margin between classes
- **MLP:** Learns complex non-linear relationships

Loss Function (Conceptual)

$$L = \frac{1}{N}\sum(y_{true} - y_{pred})^2$$

The goal is to minimize classification error and maximize generalization performance.

LLM Modelling

Instead of training, LLMs use:

$$y = LLM(prompt, text)$$

This introduces a new paradigm where knowledge is encoded in the model rather than learned from training data.

3.4 Tools and Techniques Used

The selection of tools and techniques is based on performance, scalability, and ease of integration.

Programming Language

- **Python** was chosen due to its strong ecosystem for machine learning and NLP.

Libraries Used

- **Pandas & NumPy:** Data handling
- **Scikit-learn:** ML models (LR, SVM, MLP)
- **Sentence Transformers:** Embedding generation (MiniLM)
- **Matplotlib/Seaborn:** Visualization

Why MiniLM?

- Lightweight and fast
- Captures semantic meaning effectively
- Suitable for large datasets

Why Classical Models?

- Interpretable
- Efficient training
- Strong baseline performance

Why LLMs?

- No training required (zero-shot)
- Useful in low-data scenarios
- Captures deep contextual understanding

Why Not Deep Fine-Tuning?

- High computational cost
- Requires large labelled datasets
- Less practical for rapid experimentation

System Advantages

- Modular architecture
- Scalable to new datasets
- Combines ML + LLM strengths

3.5 Detailed Technology Stack

The technology stack of the proposed system is designed to be modular, reproducible, and suitable for experimentation. Python is used as the primary programming language because it provides mature libraries for data processing, machine learning, natural language processing, and visualization. Jupyter Notebook is used during experimentation because it allows each stage of the pipeline to be executed, inspected, and documented step by step. This is helpful in research work where preprocessing choices, model parameters, and evaluation results must be reviewed repeatedly.

The data engineering layer is built using Pandas and NumPy. Pandas is used for loading CSV files, selecting relevant columns, handling labels, removing duplicate records, and organizing intermediate outputs. NumPy is used for efficient numerical operations on embedding arrays. Together, these libraries provide a stable foundation for converting unstructured text data into structured inputs that can be used by machine learning models.

The preprocessing layer performs cleaning and normalization. It removes URLs, unwanted symbols, repeated spaces, and irrelevant text fragments. The pipeline also handles empty posts and inconsistent formatting. This layer is implemented separately from model training so that the same cleaned data can be reused across classical machine learning and LLM-based experiments. Such separation improves reproducibility and makes it easier to debug errors.

The modelling layer is implemented using Scikit-learn. Logistic Regression, SVM, and MLP are trained using the same embedding features. Scikit-learn provides consistent interfaces for fitting models, predicting labels, and calculating evaluation metrics. It also supports stratified cross-validation, which is important when working with balanced but limited samples.

3.5.1 Sentence Transformer and MiniLM Module

The embedding module uses the all-MiniLM-L6-v2 sentence transformer model. Each post is tokenized and passed through the transformer network, after which pooling is used to obtain a fixed-size sentence vector. This vector represents the semantic meaning of the complete post. Because every post is converted into the same dimensional representation, the downstream classifiers can process short and long posts using a uniform input format.

MiniLM is selected because it provides a strong balance between accuracy and efficiency. Larger models may capture more context but require more memory, longer inference time, and sometimes

GPU acceleration. In contrast, MiniLM can run efficiently in a CPU-based environment while still capturing meaningful contextual information. This matches the project requirement of building a practical and low-cost detection system.

Embedding reuse is another important advantage. Once vectors are generated, they can be saved and reused for multiple experiments. This reduces repeated computation and allows fast comparison of classifiers. For example, the same embedding matrix can be used to train Logistic Regression, SVM, and MLP, and can also be used for cross-domain experiments. This design improves both speed and consistency.

3.5.2 Classical Classifier Selection

Logistic Regression is used as a baseline model because it is simple, fast, and interpretable. It helps determine whether the embedding features are linearly separable. SVM is selected because it is effective for high-dimensional feature spaces and can create a strong separating margin between classes. MLP is included to test whether a non-linear classifier can capture additional relationships in the embedding space. The combined use of these models gives a wider understanding of how semantic embeddings behave with different decision functions.

3.6 Expanded Methodological Workflow

The methodology begins with data audit and label preparation. Before training, the dataset is checked for missing text, duplicate posts, inconsistent labels, and extreme length variations. Posts are mapped into the required classes, such as mental health risk and high risk. This step is necessary because inconsistent labels can mislead the classifier and reduce the reliability of the evaluation.

After preprocessing, the cleaned text is passed to the embedding generator. The output is an embedding matrix where each row corresponds to one Reddit post. This matrix becomes the main input for the classical models. The dataset is then divided using stratified cross-validation so that each fold maintains the class distribution. This prevents evaluation bias that may occur if one fold contains too many examples of only one class.

During training, each model learns the relationship between embeddings and class labels. Logistic Regression learns a linear boundary, SVM maximizes the separation margin, and MLP learns non-linear transformations through hidden layers. The models are evaluated using accuracy, precision, recall, and weighted F1-score. Weighted F1 is especially useful because it considers both class-wise performance and class distribution.

The LLM-based workflow follows a different process. Instead of training on embeddings, the model receives the post and a task description in the form of a prompt. In zero-shot classification, no examples are provided. In few-shot classification, a small number of labelled examples are included in the prompt to guide the model. The predictions are then compared with classical model outputs to understand differences in reasoning and stability.

3.7 Utility Work and Practical Deployment Flow

The utility work of the system can be described as a set of functional modules that would support real-world use. The first module is input collection, where posts or text entries are received from a dataset or from a controlled application interface. The second module is preprocessing, where the raw input is cleaned and standardized. The third module is inference, where embeddings are generated and the trained classifier predicts the risk category.

The fourth module is risk flagging. Instead of presenting the output as a medical diagnosis, the system can mark a post as requiring attention. The fifth module is review support, where flagged cases can be inspected by a human moderator, counsellor, or domain expert. This human-in-the-loop design is important because automated models can make mistakes, especially in ambiguous or high-risk situations. The system therefore supports decision-making rather than replacing professional judgement.

For practical deployment, the model output may be combined with confidence scores, timestamps, and anonymized identifiers. A simple dashboard could display the number of flagged posts, distribution of risk categories, and examples requiring review. However, any such deployment must follow strict privacy rules and should avoid storing unnecessary personal information. Only the minimum required data should be processed.

3.8 Reliability and Safety Methodology

Reliability is addressed through cross-validation, cross-domain testing, prompt sensitivity analysis, and error analysis. These methods examine not only whether the system performs well, but also where it fails. Safety is addressed by treating predictions as assistive indicators, by emphasizing recall for high-risk cases, and by recommending human review before action. This methodology is suitable for mental health applications because it balances automation with caution.

CHAPTER 4: Implementation and Testing

4.1 Development Details

The system was developed as a modular pipeline implemented using Python and Jupyter Notebooks. The development process was divided into multiple stages corresponding to different experimental components.

Step-by-step Execution

1. **Data Loading**
    - Reddit dataset loaded from CSV files
    - Relevant columns (text, subreddit) extracted
2. **Data Preprocessing**
    - Removal of URLs, mentions, and special characters
    - Lowercasing and whitespace normalization
    - Filtering empty or irrelevant posts
3. **Label Assignment**
    - Posts categorized into:
        - Mental Health Risk
        - High Risk (Suicidal)
4. **Embedding Generation**
    - Sentence embeddings generated using MiniLM
    - Each post converted into a dense vector representation
5. **Model Training**
    - Logistic Regression, SVM, and MLP trained
    - Stratified cross-validation applied
6. **LLM Setup**
    - Zero-shot classification using BART (facebook/bart-large-mnli)
    - Prompt engineering applied for different templates
7. **Evaluation**
    - Metrics computed: Accuracy, Precision, Recall, F1-score
    - Confusion matrices generated
8. **Error Analysis**

- Predictions compared across models
- Categories identified (agreement, LLM advantage, etc.)

4.2 Experimental Setup

Hardware Configuration

- Processor: Standard CPU (Intel/AMD)
- RAM: 8–16 GB
- GPU: Not required (CPU-based execution)

Software Requirements

- Python (3.x)
- Jupyter Notebook

Libraries Used

- Pandas, NumPy (data processing)
- Scikit-learn (ML models)
- Sentence Transformers (MiniLM embeddings)
- Transformers (LLM inference)
- Matplotlib, Seaborn (visualization)

Model Configuration

- Embedding Model: all-MiniLM-L6-v2
- LLM: facebook/bart-large-mnli
- Sample Size: 200 (balanced dataset)

4.3 Test Cases

Sample Test Case Table

Input Text	Expected Output	Model Output	Result
"I feel very lonely and lost"	Mental Health Risk	Mental Health Risk	Correct
"I want to end everything"	High Risk	High Risk	Correct
"I am tired but managing"	Mental Health Risk	High Risk	Incorrect
"Nothing matters anymore"	High Risk	Mental Health Risk	Incorrect

Table 4.1. Test Cases

Validation Approach

- Stratified cross-validation used
- Balanced dataset ensures fair evaluation
- Predictions compared across ML and LLM models

4.4 Safety Measures

The system incorporates several safety and ethical considerations:

Data Privacy

- Only publicly available Reddit data used
- No personal identification included

Ethical Use

- System designed for **assistive analysis**, not diagnosis
- Results should be interpreted cautiously

Bias Handling

- Balanced dataset used to reduce bias
- Cross-domain testing ensures generalization

Security Measures

- No external API dependency (local processing)
- Safe handling of sensitive textual data

Limitations Awareness

- Model may misclassify critical cases
- Not a replacement for professional diagnosis

4.5 Utility Implementation Details

The implementation can be organized into reusable utility functions so that the experiment remains easy to maintain. A data loading utility reads the input CSV file and validates whether the required columns are present. A cleaning utility applies text normalization and returns the processed version of each post. A label mapping utility converts subreddit or dataset labels into the classes required for binary classification. Keeping these functions separate reduces code repetition and makes the pipeline easier to test.

The embedding utility is responsible for loading the MiniLM model and generating sentence vectors in batches. Batch processing is useful because it reduces memory pressure and improves execution speed. Generated embeddings can also be cached in a file so that the model does not need to recompute them every time the notebook is executed. This is especially helpful during repeated experiments with different classifiers or evaluation settings.

The training utility receives the embedding matrix and labels, applies stratified cross-validation, trains the selected model, and returns predictions with evaluation scores. The same function structure can be used for Logistic Regression, SVM, and MLP by passing model-specific parameters. This design ensures that all classical models are evaluated under similar conditions.

The LLM utility handles prompt creation and zero-shot or few-shot inference. Different prompt templates can be stored and reused, allowing controlled prompt sensitivity experiments. The output of the LLM is then mapped to the same label format as the classical models. This makes direct comparison possible and supports the unified evaluation approach of the project.

4.6 Additional Testing and Maintenance

Testing is not limited to the final classification result. The preprocessing function must be tested to confirm that it removes unwanted noise without deleting meaningful content. The embedding function must be tested to ensure that every input post produces a valid vector of the expected dimension. The training function must be tested to confirm that class labels and predictions are aligned correctly. These checks reduce implementation errors that can otherwise affect the final metrics.

Integration testing is also important. A complete sample post should pass through preprocessing, embedding generation, model prediction, and output reporting without failure. Edge cases such as empty posts, very short posts, very long posts, and posts containing only symbols should be handled safely. For mental health text, it is also important that direct self-harm expressions are not accidentally removed during cleaning.

Prompt regression testing can be used for the LLM component. A small fixed set of examples is evaluated whenever the prompt template is modified. If a new prompt greatly reduces recall or produces inconsistent labels, it should not be used for final evaluation. This supports more stable comparison between prompt designs.

4.7 Reproducibility and Documentation

Reproducibility is maintained by documenting dataset preparation, preprocessing rules, embedding model name, classifier parameters, and evaluation procedure. Random seeds should be fixed wherever possible. Intermediate files such as cleaned text and embeddings may be saved to allow verification of each stage. Proper documentation makes the project easier to extend in future work, such as adding new datasets, testing more transformer models, or developing a simple monitoring interface.

CHAPTER 5: Result and Discussion

5.1 Result Analysis

The proposed system was evaluated using both classical machine learning models and large language model (LLM)-based approaches. The results demonstrate that semantic embeddings combined with classical classifiers provide strong and stable performance for mental health risk detection.

Model	Accuracy (%)	Precision	Recall	Weighted F1	Key Insight
Logistic Regression	~76	Moderate	Moderate	~0.75	Reliable baseline
MLP	~77	Good	Good	~0.76	Better non-linearity
SVM	~79	High	High	~0/79	Best Performer

Table 5.1 Classical Model Performance

From Table 5.1, it can be observed that all three classical models achieved comparable performance, with SVM performing the best (~79% accuracy, ~0.79 F1-score). This indicates that the embedding representation effectively captures relevant linguistic patterns across models. Logistic Regression provided a reliable baseline, while MLP showed improved performance due to its ability to model non-linear relationships.

Model	DAIC → Reddit	Reddit → DAIC	Insight
Logistic Regression	0.50	0.56	Moderate
MLP	0.50	0.58	Best generalization
SVM	0.33	0.57	Weak transfer

Table 5.2 Cross-Domain Performance

Cross-domain results (Table 5.2) highlight the challenges of generalization. Models trained on Reddit performed better when transferred to other datasets compared to the reverse direction. MLP showed the best cross-domain performance, indicating better adaptability, while SVM struggled due to overfitting to dataset-specific patterns.

Model Category	**Model**	**Weighted F1**	**Observation**
Classical ML	SVM	~0.79	Best Overall
Classical ML	MLP	~0.76	Strong
Classical ML	LR	~0.75	Baseline
LLM	Zero-shot	~0.70	No training required
LLM	Few-shot	~0.60	Less Stable
LLM	BART Zero-shot	~0.55	Weak

Table 5.3: ML vs LLM Performance

The comparison between classical models and LLM-based approaches (Table 5.3) shows that trained models significantly outperform zero-shot and few-shot methods. However, LLMs still achieve reasonable performance without any training, demonstrating their potential in low-data scenarios.

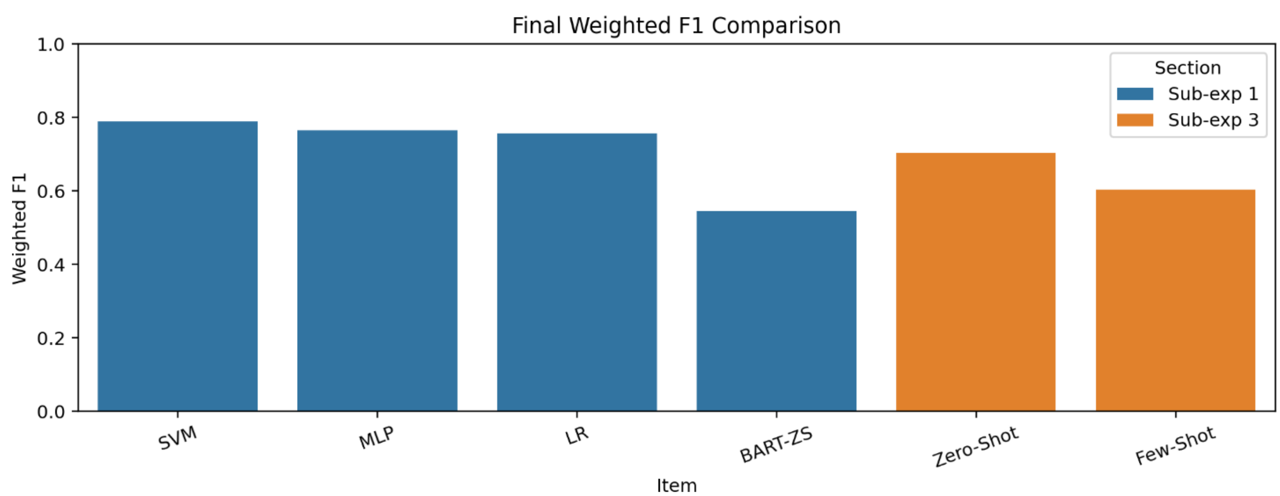

Fig 5.1 Final weighted F1 comparison

Prompt	**Accuracy**	**Weighted F1**	**Performance**
Prompt A	~0.66	~0.66	Moderate
Prompt B	~0.71	~0.70	Best
Prompt C	~0.41	~0.41	Poor

Table 5.4: Prompt Sensitivity Analysis

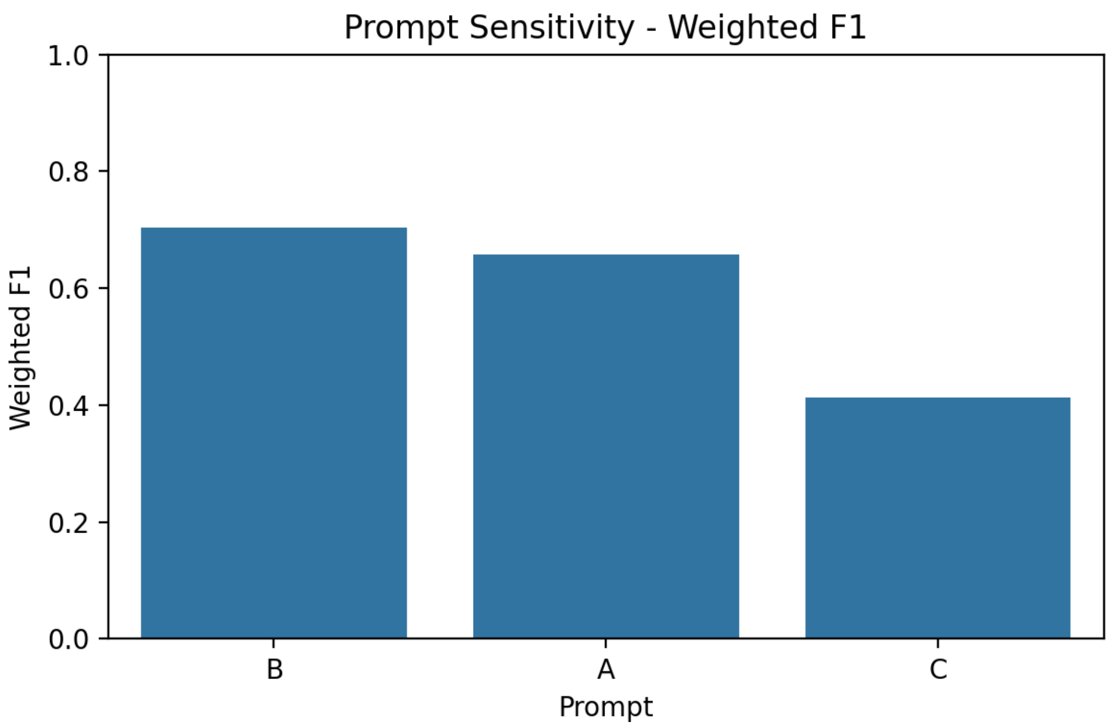

Fig 5.2 Prompt Sensitivity - Weighted F1

Prompt sensitivity analysis (Table 5.4) reveals that LLM performance is highly dependent on prompt design. Prompt B achieved the best performance (~0.70 F1), while poorly structured prompts significantly reduced accuracy. This confirms that prompt engineering is a critical factor in LLM-based classification.

Metric	**Value**
Accuracy	0.565
Precision (Weighted)	~0.58
Recall (Weighted)	0.565
Weighted F1	0.546

Table 5.5: Zero-shot Performance Metrics

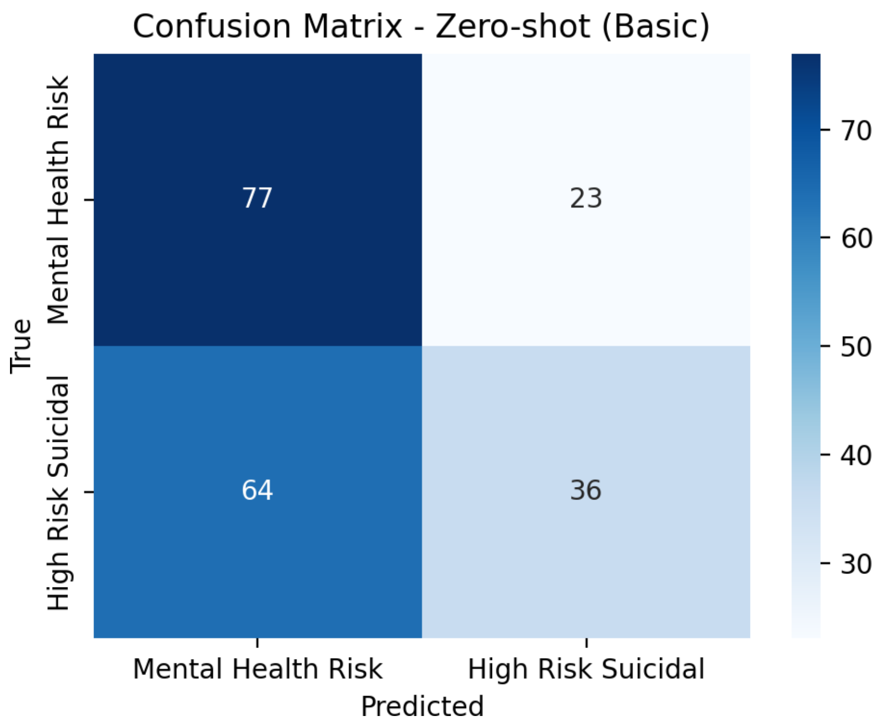

Fig 5.3 Confusion matrix – Zero-shot(basic)

Zero-shot evaluation metrics (Table 5.5) and confusion matrix analysis indicate a bias toward predicting the majority class, leading to higher false negatives in high-risk cases. This highlights a key limitation of LLM-based approaches in safety-critical applications.

Category	**Percentages**	**Interpretation**
Agreement Correct	~52%	Stable predictions
LLM Advantage	~25%	Better contextual reasoning
Classical Advantage	~17%	Better pattern detection
Hard Cases	~5%	Ambiguity

Table 5.6: Error Analysis Distribution

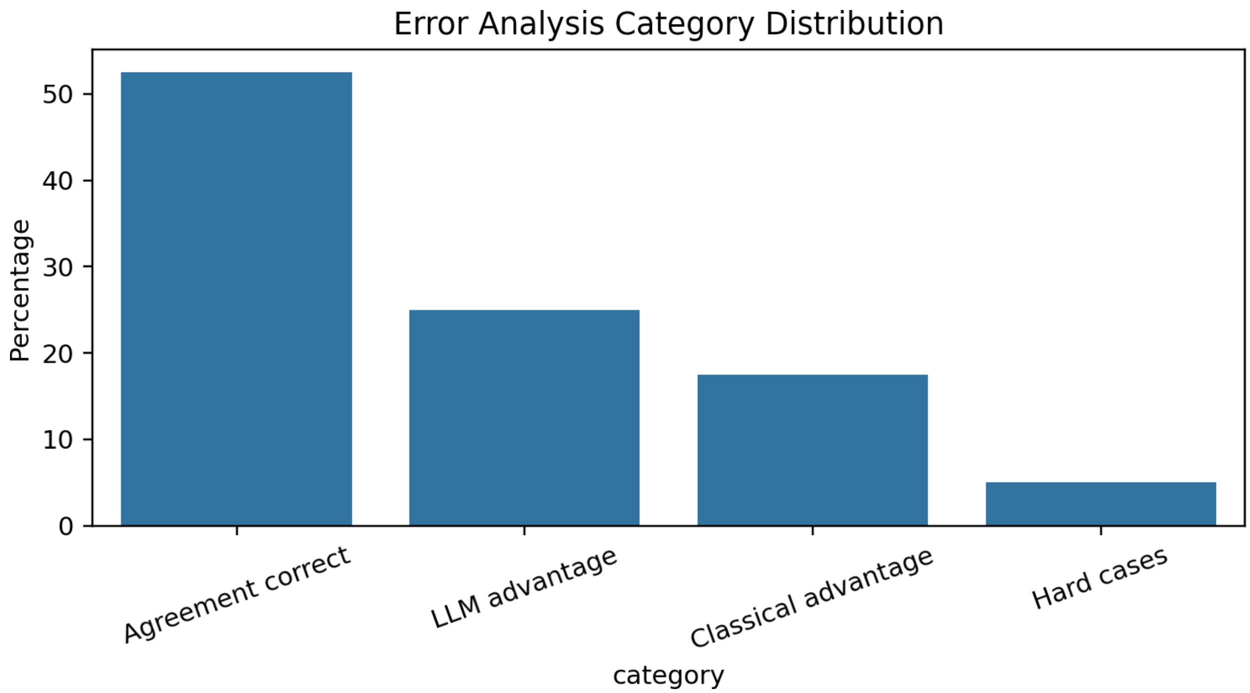

Fig 5.4 Error Analysis

Error analysis (Table 5.6) shows that more than half of the predictions are consistent across models, while LLMs perform better in context-heavy cases and classical models excel in structured patterns. A small percentage of cases remain inherently ambiguous.

5.2 <u>Comparison (with Objectives)</u>

The results align closely with the objectives defined in Chapter 1. The system successfully implements a machine learning-based framework for mental health risk detection and demonstrates the effectiveness of semantic embeddings in capturing contextual meaning from text.

The comparison between classical and LLM-based approaches provides valuable insights into their respective strengths. Classical models achieved higher accuracy and reliability due to supervised training, while LLMs demonstrated flexibility and the ability to perform classification without explicit training.

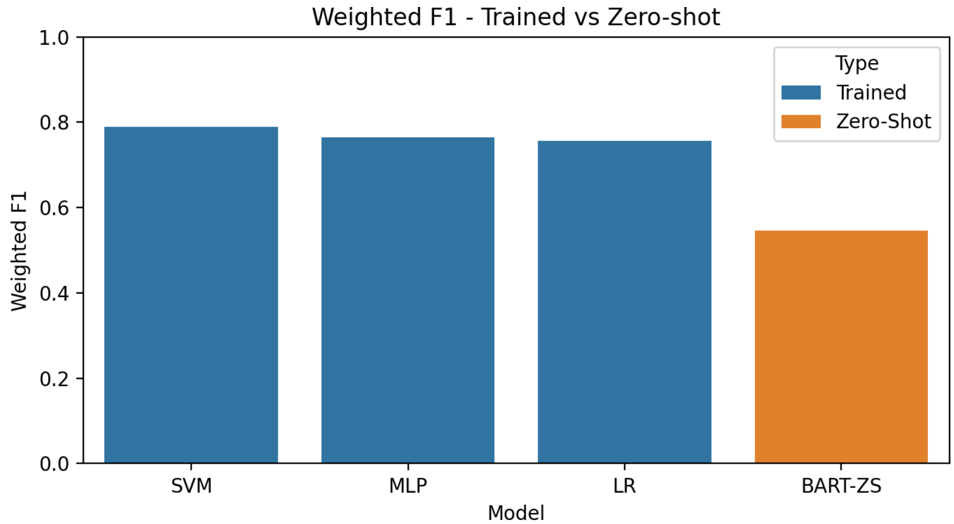

Fig 5.5 Trained vs Zero-shot

The comparison between classical and LLM-based approaches provides valuable insights into their respective strengths. Classical models achieved higher accuracy and reliability due to supervised training, while LLMs demonstrated flexibility and the ability to perform classification without explicit training.

Additionally, the inclusion of cross-domain evaluation validates the system's ability to generalize across datasets, although performance degradation highlights the need for domain adaptation techniques. Overall, the project successfully meets its objectives by providing a comprehensive evaluation of different approaches within a unified framework.

Conclusion

- All objectives were successfully achieved.
- The system provides both **high accuracy (ML)** and **flexibility (LLM)**.

5.3 <u>Economic Analysis</u>

The proposed system is highly cost-effective due to its reliance on open-source tools and lightweight computational requirements. All components of the system, including data processing, embedding generation, and model training, are implemented using freely available libraries such as Python, Scikit-learn, and Sentence Transformers.

Unlike deep learning models that require high-end GPUs and large datasets, the system operates efficiently on standard CPU-based hardware. The use of pre-trained embeddings eliminates the need for expensive training processes, further reducing computational cost.

The LLM component is implemented using locally available models, avoiding dependency on paid APIs and ensuring zero operational cost. This makes the system suitable for deployment in resource-constrained environments such as educational institutions or small-scale research setups.
Overall, the system demonstrates a favourable cost-benefit ratio, providing strong performance with minimal resource utilization.

5.4 Comparison with existing solutions

Existing mental health detection systems primarily rely on either traditional machine learning techniques or deep learning models. Keyword-based and sentiment analysis approaches are simple but lack contextual understanding, leading to lower accuracy. Traditional machine learning models using TF-IDF features improve performance but fail to capture semantic relationships.

Approach	**Strength**	**Limitation**	**Our Improvement**
Keyword-based	Simple	Low Accuracy	Better embeddings
Traditional ML(TF-IDF)	Fast	Poor Context	Semantic embeddings
Deep learning	High Accuracy	High Cost	Lightweight approach
LLM-only	Flexible	Lower Accuracy	Hybrid comparison

Table 5.7 Comparison

Deep learning and transformer-based models provide better contextual understanding but require large datasets and high computational resources, limiting their scalability. On the other hand, LLM-based approaches offer flexibility and require no training but may suffer from lower accuracy and sensitivity to prompt design.

The proposed system addresses these limitations by integrating both classical machine learning and LLM-based approaches within a single framework. By using sentence embeddings, the system captures semantic information effectively while maintaining computational efficiency.

Furthermore, the inclusion of cross-domain evaluation and error analysis provides a deeper understanding of model behaviour, which is often missing in existing systems. This makes the proposed approach more practical, scalable, and suitable for real-world mental health analysis.

Chapter 6 Conclusion and Future Scope

6.1 Summary of Achievements

The project was able to build an end-to-end textual social media-based mental health risk detection system using textual data. Its architecture incorporates various modules, such as text preprocessing, semantic embedding generation based on MiniLM, and classification based on traditional machine learning models, which include Logistic Regression, Support Vector Machines (SVM) and Multi-Layer Perceptron (MLP).

The experiment findings indicate that classical models, especially SVM, have a high-performance accuracy and F1-score. Also, the project investigated large language model (LLM)-based methods, such as zero-shot and few-shot learning, which allow classification to be performed without direct training. A comprehensive comparison of trained models and the methods based on LLM was made that can give useful insights into their weaknesses and strengths.

Cross-domain evaluation and error analysis is also introduced into the system, guaranteeing robustness and more detailed knowledge of the model behavior. Generally, the project fulfills its goal of creating a scalable and efficient system of mental health risk detection.

6.2 Effect on the Society, Sustainability and Ethical considerations.

The system suggested can make a considerable contribution to the sphere of mental health monitoring. It can help in detecting persons in distress earlier by allowing textual data to be automatically analyzed and identify possible mental health risks. This can help the professionals, researchers and support systems of the healthcare to intervene at the right time.

In terms of sustainability, the system is computedally efficient, based on lightweight models and pre-trained embeddings. This is less power intensive than deep learning systems that are large-scale, which makes it appropriate to be deployed in resource-constrained systems.

In this area, ethical considerations are paramount. The system utilizes publicly available data and does not provide any personal identifiable information, keeping users private. The system should however be used to produce predictions, which should be viewed as supportive indicators, but not clinical diagnoses. The chances of misclassification particularly false negativity in high-risk cases underscore the significance of human control.

6.3 Limitations

Although the system has shown good results, it has a number of limitations. To begin with, the dataset is restricted to textual data of a particular online platform and might not fully represent diverse populations or communication styles. This influences the generalization ability of the models, which is seen in cross domain experiments.

Second, the approaches based on LLM are very sensitive to immediate design, resulting in inconsistent performance. In particular, few-shot methods performed inconsistently because there could be bias caused by the choice of examples.

Third, the system is limited to text data and does not use multimodal information like audio or behavioral cues, which may give further context to mental health analysis.

Lastly, although classical models are effective, they use labelled data, which is not always present in the real world.

6.4 Future Work

The system can be improved in future work in a variety of ways to make it more efficient and applicable. A possible direction is the use of more sophisticated transformer-based models like BERT or RoBERTa that can enhance the contextual comprehension and classification rates.

The other notable addition is the addition of multimodal data, including speech patterns, facial expressions, or user interaction behavior, to enable a more holistic evaluation of mental health.

Future research could also aim at enhancing LLM-based methods by better prompt engineering strategies, automatic prompt optimization or domain-specific fine-tuning. Secondly, the domain adaptation methods may be considered to enhance the cross-dataset generalization.

Lastly, the system can be integrated to an active monitoring application, which can analyze the incoming data in social media and flag high-risk posts. Responsible deployment will be necessary by incorporating ethical safeguards and human-in-the-loop systems.

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